No: Date: / /

Research Project with log entity Forum

project Id : R-26-DS-011
Student Id : IT22504418
Name : A. N Sirisardena
Individual Component : Multi-Source Incident fusion, UIRs & Real Time Dashboard

2025/12/03 :- Meet our research project supervisor and discuss about few topics to do our research

2025/12/08 :- Meet our research project supervisor and discuss about few topics to do our research

2025/12/10 :- Finalized our research topic named 'AI Based emergency call and sms triage system for disaster response in Sri Lankan low resource local languages' as the research topic.

2025/12/18 :- meet our research supervisor and discussed our components with other members

2025/12/20 :- Finalized research topics with components

2025/12/21 :- register our research group and topic with relevant components.

Student Progressing well in research project Yes / NO
Signature :-

No: Date: __/__/____

Project Id :- R-26-DS-011

Student Id :- IT22504418

Name :- A.N Siriwardena

Individual Component :- Multi-Source Incident Fusion, UIRs & Real Time Dashboard

01/01/2026 :- Identified the general research domain and discussed the real world importance of an AI-based emergency call and SMS triage system for disaster response in Sri Lanka, especially for low resource local languages such as Sinhala and Tamil

04/01/2026 :- Researched how emergency communication systems work and focused on the challenges in processing multilingual emergency calls and SMS messages in Sinhala, Tamil and mixed language inputs

07/01/2026 :- Collected related research papers and reviewed previous studies connected to speech, text and emergency message analysis

08/01/2026 :- Visited DMC for the first time and discussed our ~~research~~ research topic. We talked about the importance of this research to DMC and discussed possible ways to collect data for our study.

10/01/2026 :- Prepared my part of the TAF document and focused on how the system can deduplicate incidents coming from different sources.

12/01/2026 :- Submitted the TAF document

Student progressing well in research project Yes / No

Signature :-

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Project Id :- R-26-DS-011

Student Id :- IT22504418

Name :- A.N Siriwardana

Individual Component :- Multi_Source Incident Fusion, UZRs & Real Time Dashboard

04/02/2026 :- Started to identify what data should be collected for my function and discussed the main inputs needed.

07/02/2026 :- Attended a meeting and discussed the function with industry related experts, Gained useful ideas about emergency communication, language challenges and possible user inputs

12/02/2026 :- Prepared the external data collection request form and submitted it to CDAP for data collection purposes.

16/02/2026 :- Arranged a meeting with group members and ~~both supervisors and presented our~~ discussed each member's ~~work~~ inputs and outputs to keep the overall system connected properly.

20/02/2026 :- Arranged a meeting with group members and ~~both~~ supervisor and presented our inputs and outputs. After the discussion, made several changes to improve the function.

Student progressing well in research project Yes / No

Signature :-

No: Date: / /

project Id : R-26-DS-011
Student Id : IT22504418
Name : A.N Sivaardena
Individual Component : Multi-Source Incident Fusion, UZRs, & Real Time Dashboard

01/03/2026 :- Started preparing the research proposal presentation and organized the main ideas related to the project topic

05/03/2026 :- Identified the structure of the individual report and presentation and selected the content related to my component

07/03/2026 :- Collected Insights from previous research papers and reviewed studies related to emergency calls, SMS analysis and local language processing.

14/03/2026 :- Finalized the research proposal report with the relevant details of the proposed system and individual component.

22/03/2026 :- Arranged a meeting with group members and supervisor to discuss each members's report and receive feedback for improvement.

24/03/2026 :- Finalized the research proposal presentation and obtained supervisor signatures for the research proposal report.

26/03/2026 :- preesented the research proposal presentation to the panel.

Student progressing well in research project Yes / No
Signature :-

No : _____ Date: __/__/____

project Id : R-26-DS-011
student Id : IT22504418
Name : A .N Siriwardana
Individual Component : Multi - Source Incident Fusion, UIRs & Real Time Dashboard.

01/04/2026 :- Arranged a meeting with supervisors and group members to discuss the research topic

04/04/2026 :- Searched for suitable models to use in my research component and gained additional knowledge about AI models related to the function.

09/04/2026 :- Started collecting data for the research

Signature – 

(Ms.Malithi Nawarathne) 18/04/2026 signed on behalf of the supervisor.

No: Date: __/__/____

project Id : R-26-DS-011
student Id : IT02504418
Name : A .N Siriwardena
Individual component : Multi-Source Incident Fusion, VIRS & Real Time Dashboard.

01/04/2026 :- Arranged a meeting with supervisors and group members to discuss the research topic

04/04/2026 :- Searched for suitable models to use in my research component and gained additional knowledge about AI models related to the function.

09/04/2026 :- Started collecting data for the research

10/04/2026 :- Collected the data needed and started working on sorting it.

11/04/2026 :- continued sorting the data to help with correct identification

15/04/2026 :- completed data sorting and started identifying incidents

20/04/2026 :- Had a team meeting to discuss the project plan and assign tasks.

24/04/2026 :- ~~started designing model building.~~ started designing ~~Notes~~ ~~the start of model building~~ similarity model using conversational cues

26/04/2026 :- Had a team meeting to discuss the progress as a team

27/04/2026 :- Had a discussion with DMC over the phone on the phases of Data Collection

Signature:

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project Id: R-26-DS-011

Student ID: IT22504418

Name: A. N Siriwardena

Individual component : Multi Source Incident Fusion, VIRS & real Time Dashboard

2026/05/01 - Implemented multilingual report normalization module. ~~normalization part and gazetteer. Tested~~ Developed a preprocessing component to standardize incoming voice and SMS reports by unifying field names, formatting time stamps, validating data ranges and handling missing values before further processing.

2026/05/02 - Developed AI based location extraction and geocoding functionality. Implemented a location resolution mechanism to identify Sinhala and Tamil place names from emergency reports and convert them into geographic coordinates using geocoding services.

2026/05/04 - Implemented semantic similarity detection using LaBSE embeddings. Integrated the LaBSE multilingual embeddings model to generate vector representation of reports and calculate semantic similarity between cross lingual emergency messages.

2026/05/05 - Developed machine learning based deduplication module Built and evaluated multiple classification models including XGBoost, RandomForest, LightGBM to identify whether two reports describe the same real world incident

2026/05/08 - Implemented urgency aware clustering mechanism. Developed a clustering pipeline that groups duplicate reports into unified incident records while applying stricter merge conditions for CRITICAL incidents to reduce false merge

2026/05/09 - Implemented operator feedback learning mechanism. Added a feedback loop where operator merge/split actions are logged as labeled training examples, enabling the similarity model to continuously improve over time through retraining

2026/05/10 - Had a team meeting to discuss the PPI plan and divide work among members

2026/05/12 - present the PPI to the panel

Signature:

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Project Id : R-26-DS-011

Student ID : IT22504418

Name : A. N Sirisardana

Individual component : Multi source Incident Fusion, UIRs & real time Dash board

13/05/2026 - Reviewed PPi feedback and identified requred changes to the propsed system architecture. Discussed alternative approaches and documented the necessary modifications to the project scope.

16/05/2026 - Conducted background research on LLM based multi agent systems and analyzed how different agents can be assigned specific responsibilities within the emergency incident management workflow

20/05/2026 - Redesigned the project workflow based on the new requirements. Prepared an updated system architecture diagram and outlined the interactions between the proposed agents

24/05/2026 - Defined the responsibilities, inputs, and outputs of each agent in the multi agent framework. Documented the overall processing pipe line and planned the implementation sequence.

30/05/2026 - Revised the project plan, updated technical documentation and prepared a detailed development roadmap for the next implementation phase.

Signature :

Project Id : R-26-DS-011
Student Id : IT22504418
Name : A. N Sirlwardena

Date: / /

Individual Component : Multi-source Incident Fusion, VIRs & Real Time Dashboard

04/06/2026 - According to the supervisor's advice change the plan and add fine tune models.

08/06/2026 - Prepared multilingual emergency report pairs and generated positive and negative examples for same incident and different incident classification.

13/06/2026 - Implemented LaBSE based multilingual semantic similarity to compare call and sms reports across Sinhala Singlish, tanglish, Tamil, English

20/06/2026 - Evaluated baseline similarity methods including TF-IDF, LaBSE and E5 embeddings for incident matching.

27/06/2026 - Designed the first report linking workflow using semantic similarity scores together with simple temporal and location-related information.

30/06/2026 - Fine tuned the LaBSE bi-encoder for incident-pair similarity and evaluated retrieval performance on validation and test data.

Signature:

Project Id : R-26-DS-011
Student Id : IT22504418
Name : A N Siriwardena

Individual component : Multi-source Incident Fusion, UIRS & Real Time Dashboard

05/07/2026 - Implemented candidate retrieval to shorlist the most similar previous reports before detailed comparison.

10/07/2026 - Introduced mDeBERTa - based cross-encoder scoring to improve pairwise incident matching beyond semantic similarity alone.

20/07/2026 - Compared LaBSE retrieval and mDeBERTa re-scoring results and analyzed difficult cases such as similar locations and multilingual wording.

25/07/2026 - Developed an initial merge / defer / separate decision policy using model cofidence and candidate scores.

30/07/2026 - Test the whole system

Signature :

Project Id : R-26-DS-011
Student Id : IT22504418
Name : A.N Siwardena

Individual Component : Multi-source Incident Fusion, UIRS & Real Time Dashboard

04/08/2026 - Added urgency information to the decision stage and designed an urgency-based cost matrix for false merge and false split errors

11/08/2026 - Evaluated cost-sensitive incident decisions using different urgency levels and merge/separate penalties.

17/08/2026 - Analyzed false merges, false splits and deferred cases to understand weaknesses in the existing report-level matching approach

23/08/2026 - Investigated place-name shortcuts and hard negatives cases where reports from nearby locations were incorrectly linked due to high semantic similarity

25/08/2026 - testing the system

28/08/2026 - testing the system.

Signature :

Project Id : R-26-DS-011
Student Id : IT23504418
Name : A. N Siriwardana

Individual Component : Multi-source Incident fusion, UIRs & Real Time Dashboard

01/09/2026 - Present PP2 to the panel

02/09/2026 - Reviewed PP2 feedback and identified required changes to the proposed system architecture.

04/09/2026 - Prepared and cleaned the incident dataset for training and evaluation. Organized reports, incident IDs, languages, locations and train/test/dev splits.

08/09/2026 - Improved step1 report understanding pipeline to standardize incoming call and sms reports and extract incident type, location, people, trapped, injured and help needed information by fine tuning the mt5-small.

12/09/2026 - Implemented question-anchored recovery and supporting models for understanding operator questions and caller answers.

16/09/2026 - Developed step2 evidence-pool generation. Presered caller statements without translation and categorized each evidence item.

19/09/2026 - Implemented step3A location-resolution mechanism using google places for suitable Latin-script locations and text fallback for native-script locations.

21/09/2026 - Developed step 3B geographic compatibility scorer and step 4 temporal compatibility models

26/09/2026 - Fine tuned LaBSE for step 5A multilingual semantic similarity and evaluated the model against baseline approaches.

30/09/2026 - fine tuned multilingual mDeBERTa for step 5B evidence level reasoning to identify corroboration, netural evidence, contradiction and count variation.

signature :

Project Id : R-26-DS-011
Student Id : IT22504418
Name : A · N Suiwardena
Individual Component : Multi Source Incident fusion, UIRs & Real Time Dashboard

01/10/2026 - Developed step 6 fusion model to combine geographic, temporal, semantic and evidence level features into a single probability of the same incident

02/10/2026 - Implemented step 7 cost aware merge/separate decision mechanism using the selected 3:1 false merge to false split cost ratio.

03/10/2026 - Implemented step 8 incident level matching, changed the candidate from a single report to a complete incident and created fused incident records with timelines and evidence trails.

04/10/2026 - Implemented and evaluated step 9 revision strategies including retrospective merge, detach and ambiguity detection. Automatic revision was not adopted because it increased incorrect grouping

05/10/2026 - Started doing final testing and writting the thesis
06/10/2026 - testing and thesis preparation
07/10/2026 - thesis preparation
08/10/2026 - thesis preparation
09/10/2026 - finalizing the thesis
10/10/2026 - Thesis submitted

Signature :